52. Bulgarian Mathematical Olympiad

Final round

1. If $x_1 = 2$ and $x_2 = x_3 = x_3 = x_5 = -1$, then all six sums of the form $x_1 + x_q + x_r$, $2 \le q < r \le 5$ are equal to 0 and therefore $n \ge 7$

Let now 7 sums of the form $x_p + x_q + x_r$, $1 \le p < q < r \le 5$, be equal to 0. Since $\frac{7.3}{5} > 4$, there exists x_i which occurs at least 5 times in these sums. We may assume that $i = 1$.

The sums of the form $x_1 + x_q + x_r$, $2 \le q < r \le 5$, are 6. Hence at most one of them is non-zero. So we may assume that

$$\begin{aligned} x_1 + x_2 + x_4 &= x_1 + x_2 + x_5 = x_1 + x_3 + x_4 \\ &= x_1 + x_3 + x_5 = x_1 + x_4 + x_5 = 0. \end{aligned}$$

It follows that $x_2 = x_3 = x_4 = x_5 = -\frac{x_1}{2}$ So all the 6 sums $x_1 + x_q + x_r$, $2 \le q < r \le 5$ equal 0. Since the 7-th sum which equals 0 has the form $x_p + x_q + x_r$, $2 \le p < q < r \le 5$, we get that $3x_p = 0$, which implies that $x_1 = x_2 = x_3 = x_4 = x_5 = 0$.

2. a) Set $\sphericalangle NPC = \varphi_1$ and $\sphericalangle MPC = \varphi_2$. Then

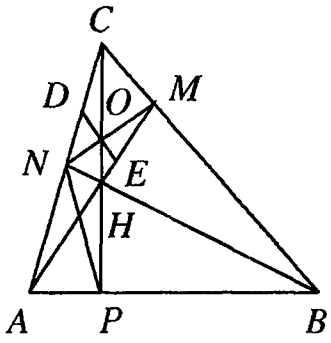
$$\frac{CN}{AN} = \frac{S_{NPC}}{S_{NPA}} = \frac{CP \sin \varphi_1}{AP \sin(90^\circ - \varphi_1)}$$

and hence

$$\tan \varphi_1 = \frac{CN.AP}{AN.CP}.$$

Analogously

$$\tan \varphi_2 = \frac{CM.BP}{BM.CP}.$$



Hence the equality $\varphi_1 = \varphi_2$ is equivalent to

$$\frac{CN.AP.BM}{AN.BP.CM} = 1$$

which follows from Ceva's theorem for $\triangle ABC$ and the lines AM, BN and CP.

b) Set $\sphericalangle NPC = \sphericalangle MPC = \varphi$, $\sphericalangle EPO = x$ and $\sphericalangle DPO = y$. It is easy to see that

$$x = y \iff \cot x = \cot y \iff \frac{\sin(\varphi - x)}{\sin x} = \frac{\sin(\varphi - y)}{\sin y}.$$

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Assume that $E \in NH$ and $D \in CM$. Then

$$\frac{ME}{EH} = \frac{S_{MEP}}{S_{HEP}} = \frac{MP\sin(\varphi - x)}{PH\sin x} \iff \frac{\sin(\varphi - x)}{\sin x} = \frac{ME.PH}{EH.MP}.$$

Similarly $\dfrac{\sin(\varphi - y)}{\sin y} = \dfrac{DN.CP}{CD.PN}$. Hence using that PO is the bisector of $\sphericalangle NPM$, i.e., $\dfrac{PM}{PN} = \dfrac{MO}{NO}$ we have to prove that

$$\frac{ME}{EH} \cdot \frac{CD}{DN} \cdot \frac{PH}{CP} \cdot \frac{NO}{MO} = 1.$$

Set $\sphericalangle NOD = \delta$ and $\sphericalangle EOP = \psi$. Then

$$\frac{ME}{EH} = \frac{S_{MEO}}{S_{HEO}} = \frac{MO\sin\delta}{OH\sin\psi} \text{ and } \frac{CD}{DN} = \frac{S_{CDO}}{S_{DNO}} = \frac{CO\sin\psi}{ON\sin\delta}.$$

Hence it remains to show that $\dfrac{OC.PH}{OH.PC} = 1$.

Applying the Menelaus theorem for $\triangle BHC$ and the line MN, $\triangle CHM$ and the line AB, and $\triangle BHM$ and the line AC, we get that

$$\frac{BN.HO.CM}{NH.OC.MB} = 1, \quad \frac{CP.HA.MB}{PH.AM.BC} = 1 \text{ and } \frac{HN.BC.MA}{BN.CM.HA} = 1.$$

Multiplying these equalities gives the desired result.

3. Let k have the given property. We have that $y_3 = 2k - 2 = 4a^2$ $(a \geq 0)$, i.e., $k = 2a^2 + 1$, Further $y_4 = 8k^2 - 20k + 13$ and $y_5 = 32k^3 - 120k^2 + 148k - 59 = 256a^6 - 96a^4 + 8a^2 + 1$.

If $a = 0$ we get that $k = 1$ and the given sequence is $1, 1, 0, 1, 1, 0, \ldots$. Hence $k = 1$ is a solution of the problem.

Let $a > 0$. Its is easy to check that

$$(16a^3 - 3a)^2 \geq y_5 = 256a^6 - 96a^4 + 8a^2 + 1 > (16a^3 - 3a - 1)^2.$$

Since y_5 is a perfect square, the first inequality must be equality, i.e., $a = 1$ and then $k = 3$.

We shall prove that $k = 3$ is a solution of the problem. In this case the sequence is defined by $y_1 = y_2 = 1$ and $y_{n+2} = 7y_{n+1} - y_n - 2$ for $n \geq 1$. Since $y_3 = 2^2$, $y_4 = 5^2$ and $y_5 = 13^2$, it is natural to conjecture that $y_n = u_{2n-3}^2$ for $n \geq 2$ where $\{u_n\}_{n=1}^{\infty}$ is the Fibonacci sequence: $u_1 = u_2 = 1$ and $u_{n+2} = u_{n+1} + u_n$ for $n \geq 1$. To prove this, we first note that $u_{n+2} = 3u_n - u_{n-2}$ and $u_{n+2}u_{n-2} - u_n^2 = 1$ for any odd $n \geq 3$. It follows that $(u_{n+2} + u_{n-2})^2 = 9u_n^2$ and thus

$$u_{n+2}^2 = 9u_n^2 - u_{n-2}^2 - 2u_{n-2}u_{n+2} = 7u_n^2 - u_{n-2}^2 - 2.$$

Hence $y_n = u_{2n-3}^2$ for $n \geq 2$.

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4. Let $A = \{a_1, a_2, \ldots, a_n\}$ be a uniform set. Set $S = a_1 + a_2 + \cdots + a_n$. It follows from the given condition that $S - a_i$ is an even number for any $i = 1, 2, \ldots, n$ Suppose that the number S is even. Then all the numbers a_i are even. Set $a_i = 2b_i$, $i = 1, 2, \ldots, n$. Then it is easy to see that the set $B = \{b_1, b_2, \ldots, b_n\}$ is uniform, too. So, we may assume that S is an odd number and whence $a_1, a_2, \ldots, a_n$ and n are also odd numbers.

We shall prove that $n = 7$. It is not difficult to check that $\{1, 3, 5, 7, 9, 11, 13\}$ is a uniform set. It remains to show that there are no unform sets with 5 elements, since it is is obvious that the sets with 3 elements are not uniform.

Suppose that $A = \{a_1, a_2, a_3, a_4, a_5\}$ is a uniform set and let $a_1 < a_2 < a_3 < a_4 < a_5$. Considering the set $A \setminus \{a_1\}$ we see that either $a_2 + a_5 = a_3 + a_4$ or $a_2 + a_3 + a_4 = a_5$. Considering the set $A \setminus \{a_2\}$ we get that either $a_1 + a_5 = a_3 + a_4$ or $a_1 + a_3 + a_4 = a_5$.

- If $a_2 + a_5 = a_3 + a_4$ and $a_1 + a_5 = a_3 + a_4$, then $a_1 = a_2$.
- If $a_2 + a_5 = a_3 + a_4$ and $a_1 + a_3 + a_4 = a_5$, then $a_1 = -a_2$
- If $a_2 + a_3 + a_4 = a_5$ and $a_1 + a_5 = a_3 + a_4$, then $a_1 = -a_2$.
- If $a_2 + a_3 + a_4 = a_5$ and $a_1 + a_3 + a_4 = a_5$, then $a_1 = a_2$.

Since all the above possibilities lead to a contradiction, we conclude that there are no unform sets with 5 elements.

5. Let $a + b + c = a^2 + b^2 + c^2 = t$. Then $t \geq 0$. On the other hand, the Root mean square – Arithmetic mean inequality implies that

$$\frac{a^2 + b^2 + c^2}{3} \geq \frac{(a + b + c)^2}{9} \iff 3t \geq t^2.$$

Hence $t \in \{0, 1, 2, 3\}$. If $t = 0$ or $t = 3$, then $a = b = c = 0$ or $a = b = c = 1$, respectively, and the the statement is trivial.

Let $t = 1$. Denote by d the product of the denominators of $|a|$, $|b|$ and $|c|$. Then $x = ad$, $y = bd$ and $z = cd$ are integers for which $x + y + z = d$ and $x^2 + y^2 + z^2 = d^2$. We may assume that $z > 0$. Note that

$$(x + y + z)^2 = x^2 + y^2 + z^2 \iff xy + yz + xz = 0 \iff (x + z)(y + z) = z^2.$$

It follows that $x + z = rp^2$, $y + z = rq^2$ and $z = |r|pq$, where p and q are coprime positive integers, and r is a non-zero integer. Since $d = x + y + z = r(p^2 + q^2) - |r|pq > 0$ we get that $r > 0$. Then

$$a = \frac{x}{d} = \frac{p(p - q)}{p^2 + q^2 - pq}, \quad b = \frac{y}{d} = \frac{q(q - p)}{p^2 + q^2 - pq}, \quad c = \frac{z}{d} = \frac{pq}{p^2 + q^2 - pq}$$

and thus $abc = \dfrac{[pq(p - q)]^2}{(pq - p^2 - q^2)^3}$.

It remains to show that $pq(p - q)$ and $p^2 + q^2 - pq$ are coprime integers. Suppose that s is a prime divisor of $pq(p - q)$ and $p^2 + q^2 - pq$. Let $s|p$. Since $s|p^2 + q^2 - pq$, then $s|q$, a contradiction. Analogously, $s \nmid q$. Hence $s|p - q$. Then $s|(p - q)^2 - (p^2 + q^2 - pq) = pq$, which is impossible.

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The case $t = 2$ can be reduced to the case $t = 1$ by setting $a_1 = 1 - a$, $b_1 = 1 - b$ and $c_1 = 1 - c$. Indeed, it is easy to check that $a_1 + b_1 + c_1 = a_1^2 + b_1^2 + c_1^2 = 1$ and $a_1 b_1 c_1 = -abc$.

6. Denote by m and a the degree and the leading coefficient of $P(x)$, respectively. Let x_n be an integer solution of the equation $P(x) = 2^n$. Since $\lim\limits_{n\to\infty} |x_n| = +\infty$, then $\lim\limits_{n\to\infty} \dfrac{a|x_n|^m}{2^n} = 1$ and hence $\lim\limits_{n\to\infty} \left|\dfrac{x_{n+1}}{x_n}\right| = \sqrt[m]{2}$.

On the other hand, $x_{n+1} - x_n$ divides $P(x_{n+1}) - P(x_n)$ and thus $|x_{n+1} - x_n| = 2^{k_n}$ for some $k_n \geq 0$. Then

$$\left|\frac{x_{n+1}}{x_n}\right| = \frac{2^{k_n}}{|x_n|} + \varepsilon_n,$$

where $\varepsilon_n = \pm 1$ and we get that

$$\sqrt[m]{2} = \lim_{n\to\infty} \left(\frac{2^{k_n}}{|x_n|} + \varepsilon_n\right) = \lim_{n\to\infty} \left(2^{k_n} \sqrt[m]{\frac{a}{2^n}} + \varepsilon_n\right).$$

Note that ε_n equals either 1 or -1 for infinitely many n. Since the two cases are similar, we shall consider only the second one. Let $1 = \varepsilon_{i_1} = \varepsilon_{i_2} = \cdots$. Then

$$\sqrt[m]{2} + 1 = \sqrt[m]{a} \lim_{j\to\infty} 2^{k_{i_j} - i_j}$$

and hence the sequence of integers $k_{i_j} - i_j$ converges to some integer ℓ. It follows that $\left(\sqrt[m]{2} + 1\right)^m = a2^{m\ell}$ is a rational number. According to the Eisenstein criteria, the polynomial $x^m - 2$ is irreducible. Hence $(x-1)^m - 2$ is the minimal polynomial of $\sqrt[m]{2}+1$. It follows that $(x-1)^m - 2 = x^m - a2^{m\ell}$ which is possible only for $m = 1$.

Let $P(x) = ax + b$. Then $a(x_2 - x_1)$ divides 2 and thus $a = \pm 1, \pm 2$. Now it follows easily that all polynomials with the desired property are of the form $P(x) = a(x + b)$, where $a = \pm 1, \pm 2$ and b is an arbitrary integer.

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